2. The Geometry of Syzygies on Stacky Curves

Let X be a projective variety and L a very ample line bundle. For the embedding $X \hookrightarrow \mathbb{P}H^0(X,L) \cong \mathbb{P}^n$ defined by L, let $S_X = S/I_X$, where $S = \mathbb{C}[x_0,\dots,x_n]$ is the standard graded polynomial ring and I_X is the homogeneous defining ideal of X. The minimal graded free resolution $0 \leftarrow S_X \xleftarrow{\epsilon} F_0 \xleftarrow{d_1} F_1 \xleftarrow{d_2} \cdots$ of S_X has terms $F_p = \bigoplus_q S(-q)^{\beta_{p,q}(X;L)}$, where $S(-q)_d = S_{d-q}$. The *graded Betti numbers (or syzygies)* $\beta_{p,q}(X;L)$ depend on the extrinsic embedding, not just on X; a central theme of the field is how they nonetheless capture the intrinsic geometry of X.

Forty years after Green's work, this connection is best understood for smooth non-hyperelliptic canonical curves of genus $g \geq 3$ ($L = K_X$). Green's conjecture [87] predicts the vanishing threshold of the quadratic strand:

$$\beta_{p,p+2}(X;K_X) = 0 \quad \Longleftrightarrow \quad p < \mathrm{Cliff}(X), \qquad 0 \leq p \leq g-3. \tag{1}$$

Here $\mathrm{Cliff}(X) := \min\{\deg(L) - 2h^0(X,L) + 2 : h^0(X,L), h^1(X,L) \geq 2\}$, with the convention $\mathrm{Cliff}(X) = 1$ for a non-hyperelliptic curve of genus 3. Voisin's breakthrough proved it for general curves of every genus [134, 135]; new proofs [4, 124, 100, 102] and refinements [76, 52, 74, 75, 66, 101] followed.

This project develops the corresponding theory for *stacky curves* [2, 44, 1], where S is nonstandard graded and its geometry is visible only through fractional divisors. The classical theory of the geometry of syzygies is a theory of standard graded rings, and it does not directly cover section rings requiring generators of different degrees. Already for a point O on an elliptic curve E, the section ring $\bigoplus_n H^0(E,nO)$ needs generators in degrees $1,2,3$ and is cut out by the Weierstrass equation $y^2 = x^3 + axz^4 + bz^6$, a sextic in $\mathbb{P}(1,2,3)$. This is not an edge case but the general picture. By Demazure–Pinkham [122, 65, 136], every finitely generated normal domain R of dimension two with $R_0 = \mathbb{C}$ and an effective positive grading is $\bigoplus_{n\geq 0} H^0(C, \lfloor nD \rfloor)$ for an ample $\mathbb{Q}$-divisor D on a curve C: a section ring on a stacky curve whose stacky points record the fractional part of D. Modular curves are flagship examples; rigidifying $X(1) \cong \mathcal{P}(4,6)$ gives $\mathcal{P}(2,3)$, whose log canonical ring (with the cusp) is $\mathbb{C}[E_4,E_6]$ in degrees $2,3$. Even-weight forms on $X_0(N)$ and Shimura curves give (log) canonical rings, with weight $2d$ in degree d. Voight and Zureick-Brown [133] proved a stacky Noether–Petri theorem presenting these rings (see also [120, 106]), but their syzygies are uncharted: stacky analogues of Green's N_p theorem and Green's conjecture are wide open. This project builds a theory in which fractional divisors play the role that degree plays classically, extending Mumford's and Green's theorems to stacky curves and answering [34, Ques. 7.7]. Results without citations in this section are work ongoing with or inspired by the PI, M. Banks, and P. Wei.

2.1 Weighted Canonical Rings A *stacky curve* $\mathcal{X}$ is a smooth proper geometrically connected Deligne–Mumford stack of dimension one with a dense open subscheme. (Throughout this section, $\mathbb{K} = \mathbb{C}$.) Concretely, $\mathcal{X}$ is a curve C with finitely many *stacky points* carrying cyclic stabilizers [133, Lem. 5.3.10], and the coarse space map $\pi\colon \mathcal{X} \to C$ forgets these stabilizers. If $x_1,\dots,x_s$ are the stacky points, $e_i = \#G(x_i)$, $P_i = \pi(x_i)$, and $g = g(C)$, the *signature* of $\mathcal{X}$ is $(g; e_1,\dots,e_s)$. Following [133], the *canonical divisor* of $\mathcal{X}$ is

$$K_{\mathcal{X}} = \pi^* K_C + \sum_i (e_i - 1)\, x_i = \pi^* D \qquad \text{where} \qquad D = K_C + \sum_i \left(1 - \tfrac{1}{e_i}\right) P_i \tag{2}$$

and its degree is $\deg(K_{\mathcal{X}}) = 2g - 2 + \sum_i (1 - 1/e_i)$, which may be a rational number. Thus the genus of $\mathcal{X}$, defined by $\deg(K_{\mathcal{X}}) = 2g_{\mathcal{X}} - 2$, is the rational number $g_{\mathcal{X}} = g + \frac{1}{2}\sum_i (1 - 1/e_i) \geq g$. We say $\mathcal{X}$ is *hyperbolic* when $\deg(K_{\mathcal{X}}) > 0$, which we assume going forward. The *canonical ring* of $\mathcal{X}$ is

$$R(\mathcal{X}, K_{\mathcal{X}}) := \bigoplus_{d\in\mathbb{Z}} H^0(\mathcal{X}, dK_{\mathcal{X}}) \cong \bigoplus_{d\in\mathbb{Z}} H^0(C, \pi_*(dK_{\mathcal{X}})) \cong \bigoplus_{d\in\mathbb{Z}} H^0(C, \lfloor dD \rfloor),$$

where the *round-down* of a $\mathbb{Q}$-divisor $D = \sum_i a_i P_i$ on C is the honest divisor $\lfloor D \rfloor := \sum_i \lfloor a_i \rfloor P_i$, and the last isomorphism follows from [133, § 5.4]. When $\mathcal{X}$ is understood, we write R for $R(\mathcal{X}, K_{\mathcal{X}})$.

This round-down is the essential difference from the classical theory. Since $\lfloor D \rfloor = K_C$, $R_1 = H^0(\mathcal{X}, K_{\mathcal{X}}) \cong H^0(C, K_C)$ has dimension $g(C)$ rather than $g_{\mathcal{X}}$. When stacky points are present, these sections vanish along the ramification divisor $K_{\mathcal{X}} - \pi^* K_C$, so they neither embed $\mathcal{X}$ nor generate R. Adding higher-degree generators yields an embedding into a weighted projective stack [26]; algebraically, R is a quotient of a polynomial ring with a nonstandard $\mathbb{Z}$-grading.

Following Brown and Erman [34, § 3.1], define the *minimal weighted canonical series* $\mathcal{W} = \bigoplus_{d \geq 1} \mathcal{W}_d$ by setting $\mathcal{W}_1 = R_1$ and, for $d \geq 2$, choosing $\mathcal{W}_d$ as a complement in R_d to the degree-d part of the subalgebra generated by $\mathcal{W}_1, \ldots, \mathcal{W}_{d-1}$. Then $\mathcal{W}$ embeds $\mathcal{X}$ into $\mathcal{P}(\mathcal{W})$, and $R \cong S/I_{\mathcal{X}}$, where $S = \operatorname{Sym}(\mathcal{W})$. For $g \geq 2$, the stacky Noether–Petri theorem gives sharp degree bounds of $e := \max(1, e_1, \ldots, e_s)$ for generators and $2e$ for relations, subject to exceptional cases [133, Cor. 1.4.2]. Their inductive presentation also describes relative generators and gives criteria for minimality [133, Thm. 8.3.1(a),(d)]. Theorem 2.1 makes the full weight multiset explicit in our setting. Let $\nu_d = \dim \mathcal{W}_d$; the *weight multiset* of $\mathcal{W}$ has ν_d copies of each weight d, written d^{ν_d}. Set $N = \sum_d \nu_d$, $\sigma = \sum_d d\,\nu_d$, and $c = N - 2$, so $\mathcal{P}(\mathcal{W}) = \mathcal{P}(1^{\nu_1}, 2^{\nu_2}, \ldots)$ has dimension $N - 1$.

Theorem 2.1. *Let $\mathcal{X}$ be a hyperbolic stacky curve with $g \geq 3$. The weight multiset of $\mathcal{W}$ is $(1^g,\ (2, 3, \ldots, e_1),\ \ldots,\ (2, 3, \ldots, e_s))$, with 2^{g-2} inserted if and only if C is hyperelliptic.*

For a fixed signature, the weights detect whether C is hyperelliptic; the generation degree is e for non-hyperelliptic C and $\max\{2, e\}$ otherwise. Riemann–Roch gives sections $y_{i,j} \in R_j$ for $2 \leq j \leq e_i$ with exactly one pole, at P_i of order $j - 1$. As every generator of degree m has pole order at most $m - 1$, these are indecomposable, and products $\omega \prod_s y_{i,j_s}$ with $\omega(P_i) \neq 0$ realize every admissible pole order. Filter R by pole order at the stacky points: $F_\ell R_d := H^0\big(C, dK_C + \sum_i \min\{\ell, b_i(d)\} P_i\big)$ with $b_i(d) = \lfloor d(e_i - 1)/e_i \rfloor$. This multiplicative exhaustive filtration has $F_0 R = R(C, K_C)$. Let $\Gamma_e = \{(n, \ell) \in \mathbb{N}^2 : 0 \leq \ell \leq \lfloor n \frac{e-1}{e} \rfloor\}$ and $T_e = \mathbb{C}[\Gamma_e]$, graded by n. The flat Rees degeneration $R \rightsquigarrow \operatorname{gr}_F R$ separates principal parts at the stacky points.

Theorem 2.2. *For $g \geq 3$, $\operatorname{gr}_F R$ is the fiber product of $R(C, K_C)$ and $\prod_i T_{e_i}$ over $\prod_i \mathbb{C}[z_i]$, via evaluation in fixed trivializations of $K_C|_{P_i}$ and projection to pole order zero;* $\deg z_i = 1$. *Thus there is an exact sequence*

$$0 \longrightarrow \operatorname{gr}_F R \longrightarrow R(C, K_C) \oplus \bigoplus_i T_{e_i} \longrightarrow \bigoplus_i \mathbb{C}[z_i] \longrightarrow 0. \tag{3}$$

Since $H^1(C, nK_C) = 0$ for $n \geq 2$, graded pieces are leading coefficients of principal parts, which multiply monomially at one point and vanish across distinct points. As the generators of Theorem 2.1 generate $\operatorname{gr}_F R$, $\beta(R)$ is $\beta(\operatorname{gr}_F R)$, computed from (3), minus consecutive cancellations.

2.2 The Shape of Stacky Green's Conjecture Write $K_{p,q} := \operatorname{Tor}_p^S(R, \mathbb{C})_{p+q}$ and $\beta_{p,p+q} = \dim K_{p,q}$ over the minimal ambient ring $S = \operatorname{Sym}(\mathcal{W})$. A non-hyperelliptic canonical curve has four Betti rows and $g - 1$ columns. The endpoint rows each contain one 1; duality exchanges the middle rows: $\beta_{p,j}(C; K_C) = \beta_{g-2-p,\, g+1-j}(C; K_C)$. Green's conjecture determines the linear-strand length and, with the Hilbert function, the full table for general C, whose middle rows share no internal degree. Stacky canonical rings remain Gorenstein but can have many more rows.

Theorem 2.3. *Let $\mathcal{X}$ be a hyperbolic stacky curve with $g \geq 3$. Let $R = S/I$ be a minimal graded presentation, where $S = \mathbb{K}[x_1, \ldots, x_N]$ has total weight σ, and let $c = N - 2$.*

(1) R is Gorenstein, with $\beta_{i,j}(R) = \beta_{c-i,\sigma+1-j}(R)$.

(2) The weighted regularity, in the sense of [34]*, of R is* 3.

(3) Its nonzero Betti entries lie in rows $0 \leq j - i \leq \rho - 1$, where $\rho = \sigma - N + 4$, and both endpoint rows are attained. By Theorem 2.1, $\rho = 4 + \sum_i \binom{e_i}{2}$ if C is non-hyperelliptic and $\rho = g + 2 + \sum_i \binom{e_i}{2}$ if C is hyperelliptic.

Parts (1) and (2) follow from a local cohomology calculation showing that R is a two-dimensional Cohen–Macaulay normal domain with $\operatorname{pd}_S R = c$, together with graded local duality, or alternatively

coarse

	0	1	2	3	4
0	1	–	–	–	–
1	–	6	8	3	–
2	–	3	8	6	–
3	–	–	–	–	1

$(6;2)$

	0	1	2	3	4	5
0	1	–	–	–	–	–
1	–	6	8	3	–	–
2	–	8	24	24	8	–
3	–	–	3	8	6	–
4	–	–	–	–	–	1

$(6;3)$

	0	1	2	3	4	5	6
0	1	–	–	–	–	–	–
1	–	6	8	3	–	–	–
2	–	8	24	24	8	–	–
3	–	6	24	36	24	6	–
4	–	–	8	24	24	8	–
5	–	–	–	3	8	6	–
6	–	–	–	–	–	–	1

FIGURE 2. Betti tables of a genus-6 trigonal canonical curve C and of stacky curves $(6;2)$, $(6;3)$ over C.

from Watanabe's canonical module formula [136]. The pattern behind part (3) is striking: over a non-hyperelliptic coarse curve, whose canonical Betti table has 4 rows and $g-1$ columns, each stacky point of order e adds $e-1$ columns and $\binom{e}{2}$ rows (Figure 2). The two consequences of classical Green's conjecture thus become separate, new problems in the stacky world.

Research Problem 2.4. *Let $\mathcal{X}$ be hyperbolic, with non-hyperelliptic coarse space C of genus $g \geq 3$.*

(1) (Weak Stacky Green's Conjecture): Is a weighted analogue of Green's N_p-condition determined by $\mathrm{Cliff}(C)$ *and the signature?*

(2) (Strong Stacky Green's Conjecture): For a sufficiently general stacky curve $\mathcal{X}$, is $\beta(\mathcal{X}; K_{\mathcal{X}})$ determined by $\mathrm{Cliff}(C)$ *and its Hilbert function?*

2.3 Toward a Weak Form of Stacky Green's Conjecture Here and in §2.4, $\mathcal{X}$ and C are as in Research Problem 2.4. The weak form uses Brown–Erman's nonstandard N_p-conditions: $Z \subset \mathbb{P}(\mathcal{W})$ is *weighted* N_p if it is normally generated and F_i is generated in degree at most w^{i+1} for all $i \leq p$, where w^j sums the j largest weights [34, Def. 1.2]; standard weights recover Green's classical N_p. Let $\tau_{\mathrm{BE}}(\mathcal{X})$ be the smallest $p \geq 1$ at which weighted N_p fails. For non-stacky curves, their theorem gives weighted N_p for $1 \leq p \leq \dim \mathcal{W} - g - 2$ for any normally generated weighted series $\mathcal{W}$ of a line bundle of degree $\geq 2g+1$ with $\nu_1 > g$ [34, Thm. 1.7]. The stacky canonical series has $\nu_1 = g$, beyond their methods; [34, Ques. 7.7] requests this canonical analogue. We predict that stacky points shift the coarse threshold by $\kappa := N - g = \sum_i (e_i - 1)$, the number of higher-weight generators. Let $\tau(C, K_C)$ be the first failure of classical N_p. By duality, $\tau(C, K_C) = g - 2 - \ell(C)$, where $\ell(C) := \max\big(\{0\} \cup \{r \geq 1 : \beta_{r,r+1}(C; K_C) \neq 0\}\big)$.

Conjecture 2.5 (Weak Stacky Green's Conjecture)**.** *The threshold is $\tau_{\mathrm{BE}}(\mathcal{X}) = \tau(C, K_C) + \kappa$; equivalently, $\tau_{\mathrm{BE}}(\mathcal{X}) = (N-2) - \ell(C)$.*

For general coarse curves, Voisin's theorem [134, 135] gives $\tau(C, K_C) = \mathrm{Cliff}(C)$, so Conjecture 2.5 predicts $\tau_{\mathrm{BE}}(\mathcal{X}) = N - g + \mathrm{Cliff}(C)$, uniformly over signatures and markings. The same shift motivates Conjectures 2.13 and 2.14. The first ingredient is the following comparison of linear strands.

Proposition 2.6. *For every $r \geq 1$, the inclusion $R(C, K_C) \hookrightarrow R$ induces an isomorphism $K_{r,1}(C, K_C) \xrightarrow{\sim} K_{r,1}(\mathcal{X}, K_{\mathcal{X}})$. In particular, $\beta_{r,r+1}(\mathcal{X}; K_{\mathcal{X}}) = \beta_{r,r+1}(C; K_C)$.*

Proposition 2.6 follows by comparing Koszul cycles and boundaries using minimality of the weighted generators. Together with duality (and the final Betti entry when $\ell(C) = 0$), it gives the nonvanishing half of Conjecture 2.5. For vanishing, let u_k be the sum of the k smallest weights. Duality reduces the problem to $\beta_{r,q}(R) = 0$ for $\ell(C) < r < c$ and $q \leq u_{r+1}$. Comparison and minimality settle $r < g$. I will begin with one order-four point, where the only new requirement is $\beta_{g,g+2} = 0$, equivalently $\beta_{1,8} = 0$. The mapping cone of (3), using the semigroup relations of T_4, will test this on the special fiber and isolate any cancellation needed after deformation. For arbitrary signatures, I will insert the weighted Eagon–Northcott resolutions of the regraded rational-normal-curve rings T_{e_i} (see [64, 12]) into the fiber-product mapping cone and bound its

surviving shifts. Showing that every surviving shift in homological degree $g \le r < c$ exceeds u_{r+1} would complete the vanishing argument, proving Conjecture 2.5.

Finally, gonality and Clifford index supply no additional stacky invariant (here $\mathrm{Cliff}(\mathcal{X})$ is defined using line bundles with $h^0, h^1 \ge 2$, with the usual low-genus conventions).

Lemma 2.7 (Brill–Noether rigidity)**.** *We have* $\mathrm{Cliff}(\mathcal{X}) = \mathrm{Cliff}(C)$ *and* $\mathrm{gon}(\mathcal{X}) = \mathrm{gon}(C)$*. Every line bundle attaining either minimum is a pullback.*

Indeed, write $\mathcal{L} = \pi^* M \otimes \bigotimes_i \mathcal{T}_i^{a_i}$, where $\mathcal{T}_i = \mathcal{O}_{\mathcal{X}}(x_i)$ and $0 \le a_i < e_i$. The round-down formula gives $\dim H^i(\mathcal{X}, \mathcal{L}) = \dim H^i(C, M)$ and $\deg_{\mathcal{X}} \mathcal{L} = \deg M + \sum_i \frac{a_i}{e_i}$. Nonzero fractional parts increase degree without changing cohomology. This proves both equalities and forces every minimizer to have trivial stacky characters. The remaining Betti rows may require finer invariants.

2.4 Toward a Strong Form of Stacky Green's Conjecture The strong form asks whether g, $\mathrm{Cliff}(C)$, and the signature determine the full Betti table for general coarse C.

Theorem 2.8 (One stacky point of order two)**.** *For* $\mathcal{X} = \sqrt[2]{(C, P)}$*, with out-of-range terms zero,*

$$\beta_{p,p+q}(R) = \beta_{p,p+q}\big(R(C, K_C)\big) + \beta_{p-1,p+q-2}\big(R(C, K_C)\big) + \delta_{q,2}\tbinom{g-1}{p} - \varepsilon_{p,q},$$

where $\varepsilon_{p,q} = 1$ *exactly for* $(p, q) \in \{(1, 1), (0, 2), (g-1, 2), (g-2, 3)\}$ *and zero otherwise.*

Thus Problem 2.4(2) has an affirmative answer here: $\beta(R)$ is independent of P. Indeed, $\mathcal{W}$ has weights $1^g, 2$, so $N = g + 1$, $c = g - 1$, and $\rho = 5$. Proposition 2.6 and Theorem 2.3 specify rows $0, 1, 3, 4$, leaving only row 2. In each degree j the Hilbert numerator, which depends only on the signature, determines the sole unknown $\beta_{j-2,j}$; for $\rho \ge 6$ this fails.

Conjecture 2.9 (Strong Stacky Green's Conjecture I)**.** *The Betti numbers* $\beta_{p,q}(\mathcal{X}; K_{\mathcal{X}})$ *depend only on* C *and the signature of* $\mathcal{X}$*, and not on the position of the stacky points.*

Proposition 2.6 and Theorem 2.3 confine positional dependence to the middle rows. I will test it through the connecting-map ranks below; rank drops would identify pointed invariants beyond the gonality and Clifford index fixed by Lemma 2.7.

Our degeneration and computations suggest the following formula. For a graded S-module M, write $B_M(t, u) = \sum_{p,j} \beta^S_{p,j}(M) t^p u^j$ and $B_{\mathcal{X}} = B_R$. Let $J \subset S$ be generated by the variables of degree ≥ 2, and let P_C be the Betti polynomial of the coarse canonical ring over S/J. Set $B_J^\vee(t, u) = t^{c-1} u^{\sigma+1} B_J(t^{-1}, u^{-1})$.

Conjecture 2.10 (Strong Stacky Green's Conjecture II)**.** *We have*

$$B_{\mathcal{X}} = B_{S/J}\left(P_C + (1+tu)^{g-1} \sum_i \frac{u^2 + \cdots + u^{e_i}}{1 + tu^{e_i}}\right) - (1+t)(B_J + B_J^\vee).$$

The first term combines the coarse resolution with the higher-weight generators and the Koszul complexes on canonical forms vanishing at each P_i. The subtracted terms record consecutive cancellations paired by Gorenstein duality. To approach this formula, I will first compute the resolution of the special fiber (3), then control its cancellations under deformation.

Research Goal 2.11. *Prove that* $B_{\mathrm{gr}_F R}$ *equals the right-hand side of Conjecture 2.10 plus* $(1+t)B_J^\vee$*.*

Semicontinuity and Gorenstein duality would then bound $\beta(R)$ by the conjectured table when $(1+t)B_J$ and $(1+t)B_J^\vee$ have disjoint support, as for $\kappa < g - 2$. In this range, proving equality reduces to excluding further cancellations under deformation. To control these cancellations geometrically, M. Banks, P. Wei, and I have developed a weighted analogue of Lazarsfeld's kernel-bundle method [108, 86], as suggested in [34, §7.3]. Set $\mathcal{L} = \mathcal{O}_{\mathcal{X}}(K_{\mathcal{X}})$ and define

$$0 \longrightarrow \mathcal{M}_{\mathcal{W}} \longrightarrow \mathcal{E}_{\mathcal{W}} \coloneqq \bigoplus_{d \ge 1} \mathcal{W}_d \otimes \mathcal{L}^{-d} \longrightarrow \mathcal{O}_{\mathcal{X}} \longrightarrow 0.$$

Exterior powers and cohomology give $\operatorname{Tor}_p^S(R,\mathbb{C})_j \cong \ker H^1\big(\wedge^{p+1}\mathcal{M}_{\mathcal{W}}\otimes\mathcal{L}^j \to \wedge^{p+1}\mathcal{E}_{\mathcal{W}}\otimes\mathcal{L}^j\big)$. I will begin with $\mathcal{X}=\sqrt[3]{(C,P)}$, the first signature beyond the order-two case, with weights $1^g,2,3$. Writing $\mathcal{T}=\mathcal{O}_{\mathcal{X}}(x)$, successive evaluation maps filter $\mathcal{M}_{\mathcal{W}}$ with factors $\pi^*M_{K_C}\otimes\mathcal{L}^{-1}$, $\mathcal{L}^{-2}\otimes\mathcal{T}^{-1}$, and $\mathcal{L}^{-3}\otimes\mathcal{T}^{-1}$. Exterior powers and pushforward reduce the calculation to twists of $\wedge^a M_{K_C}$ by powers of K_C and multiples of P. The two extensions defining the filtration need not split; I will compute their connecting maps and combine them with the map to $\wedge^{p+1}\mathcal{E}_{\mathcal{W}}$ in the Tor formula. Here only rows $2,3,4$ remain undetermined, and duality exchanges rows 2 and 4, leaving $\beta_{p,p+2}$ and $\beta_{p,p+3}$ as the independent targets. For arbitrary signatures, I will express the differentials governing cancellation under the pole degeneration through the evaluation filtration and determine their ranks as the markings vary. Proving the prescribed ranks and ruling out additional cancellations would establish the full Betti formula for general markings; the loci where these ranks change would then show whether the formula extends to every marking or requires additional pointed invariants.

Schreyer's early work on Green's conjecture [125] remains, to my mind, a lovely demonstration of computation laying bare the geometry of syzygies. In that spirit, I propose to push further the computational tools that led to the conjectures and partial results above.

Research Goal 2.12. *Determine the generic Betti tables for signatures* $(g;3)$*,* $(g;4)$*, and* $(g;2,2)$ *with* $3\le g\le 6$*, letting computation test and guide Conjectures 2.9 and 2.10.*

2.5 Beyond Stacky Canonical Curves The canonical case suggests a broader principle: coarse geometry controls syzygetic thresholds, while stacky points shift their homological position. I will test this for general section rings, addressing Brown and Erman's question on weighted N_p [34, Ques. 7.6]. Let $\mathcal{D}=\pi^*L\otimes\bigotimes_i\mathcal{T}_i^{a_i}$, where L is very ample on C, $0<a_i<e_i$, $\gcd(a_i,e_i)=1$, and $\mathcal{T}_i=\mathcal{O}_{\mathcal{X}}(x_i)$ as above, so $\deg\mathcal{T}_i=1/e_i$, $\mathcal{T}_i^{e_i}\cong\pi^*\mathcal{O}_C(P_i)$, and $\lfloor\mathcal{D}\rfloor=L$. When $\deg L\ge 2g-1$, the coarse H^1 vanishings extend the pole filtration of Theorem 2.2 to $R(\mathcal{X},\mathcal{D})$, replacing $b_i(d)$ by $\lfloor da_i/e_i\rfloor$ and T_{e_i} by the semigroup ring of $\{(n,\ell):0\le\ell\le\lfloor na_i/e_i\rfloor\}$.

The first regime is $\deg L\ge 2g+1$, where classical vanishing theorems [87, 86] provide a starting point. Let $\mathcal{W}$ minimally generate $R(\mathcal{X},\mathcal{D})$ and set $\kappa=\dim\mathcal{W}-h^0(C,L)$, the number of generators of weight greater than one. This depends on the stacky twist: $\kappa=\sum_i(e_i-1)$ when every $a_i=e_i-1$, but $\kappa=s$ when every $a_i=1$. I predict that these generators shift the coarse threshold without otherwise changing it. As in §2.3, $\tau_{\mathrm{BE}}(\mathcal{X},\mathcal{D})$ and $\tau(C,L)$ denote the first failures of weighted and ordinary N_p, respectively, with either threshold equal to ∞ if no failure occurs.

Conjecture 2.13. *If* $\deg L\ge 2g+1$*, then* $\tau_{\mathrm{BE}}(\mathcal{X},\mathcal{D})=\tau(C,L)+\kappa$.

I will begin with one stacky point of order e and $\mathcal{D}=\pi^*L\otimes\mathcal{T}$, so $\kappa=1$. Write $R=R(\mathcal{X},\mathcal{D})$. The evaluation filtration has factors $\pi^*M_L\otimes\mathcal{D}^{-1}$ and $\mathcal{D}^{-e}\otimes\mathcal{T}^{-1}$. In the first forbidden degree $j=p+e+1$, exterior powers and pushforward give $\wedge^{p+1}M_L\otimes L^e(P)$ and $\wedge^p M_L\otimes L$. Coarse regularity kills the first H^1, the target H^1 in the Tor formula, and the terms in higher forbidden degrees; the second is $K_{p-1,2}(C,L)$. The first target is therefore the comparison $\operatorname{Tor}_p^S(R,\mathbb{C})_{p+e+1}\cong K_{p-1,2}(C,L)$, proving the shift by one. For general κ, I will track the vanishing orders removed by successive evaluation maps and show that only the factor containing all higher-weight generators can contribute in forbidden degrees. This would recover the coarse Koszul group and explain the full shift. I will then test the method on log canonical rings, including modular-form rings with cusps, without using Gorenstein duality. This complements recent work in nonstandard gradings [32, 31, 11, 114, 63, 37].

At degree $2g$, secant geometry gives a sharper target. For $g=2n+1$ and $\operatorname{Cliff}(C)=n$, Farkas–Kemeny show that a very ample, normally generated line bundle L of degree $2g$ satisfies N_{n-1} if and only if it is *n-very ample*: every subscheme of length $n+1$ imposes independent conditions [74, Thm. 1.4]; cf. [73]. I predict that this criterion, shifted by κ, persists for general stacky points.

Conjecture 2.14 (Stacky Secant Conjecture)**.** *For C and L as above, fixed $e_i\ge 2$, and general distinct $P_i\in C$, set $\mathcal{D}=\pi^*L\otimes\bigotimes_i\mathcal{T}_i^{e_i-1}$, $\kappa=\sum_i(e_i-1)$. Then $(\mathcal{X},\mathcal{D})$ satifies weighted $N_{n-1+\kappa}$ if and only if L is n-very ample.*

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